

Section 7.3: Measurement Applications (Data Logging)

Computers are sometimes used alongside **sensors** to **measure quantities** such as **pressure**, **temperature**, **radiation** etc. This is known as '**data logging**'.

Real-life examples that we will discuss include:

- 1 Scientific Experiments**
- 2 Electronic Timing**
- 3 Environmental Monitoring**

Key Concepts of this section:

- # Understand how **sensors** can be used to gather **measurable data**.
- # Know why computers can **log and measure data** more effectively than humans.
- # Be able to discuss examples of how **computers and sensors** can be used to **measure data**.

Sensors and Data Logging

What are sensors used for?

- # Sensors are used to **automatically detect and measure physical quantities**.
- # If the sensor is **connected to a computer**, the computer can read the data coming from the sensor and **log it at pre-set time intervals** (hourly for example). This is sometimes called '**data logging**'.
- # This allows us to see **changes** in the quantities (temperature increasing or decreasing for example)



Key Words:

Sensors, Data Logging,
Automatic, Measure

Examples of Sensors

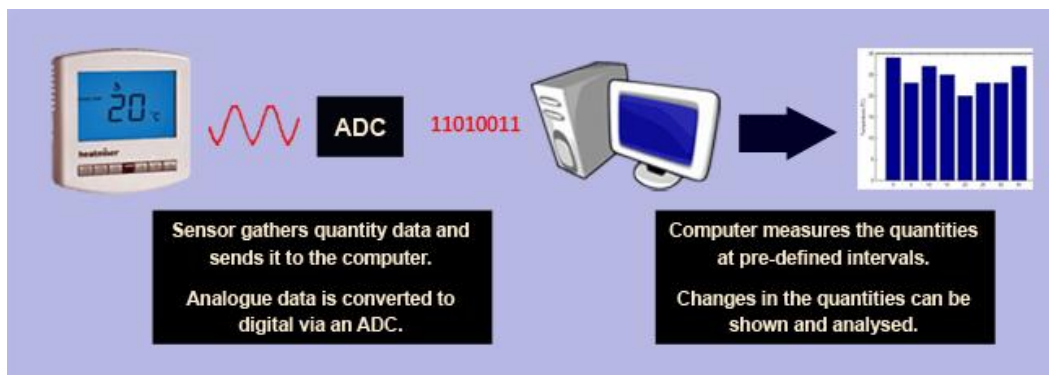
Temperature/Heat - Detect changes in temperature.

Light Sensors - Used to detect the brightness of light.

Sound Sensors - Measure the loudness of sound.

Humidity Sensors - Detect air moisture.

Infrared Sensors - Detect movement



(Used in burglar alarms).

NOTE!

Most sensors read data in **analogue format**. Before the data can be used in a digital computer it must be converted using an **analogue to digital converter (ADC)**

Why use a computer to measure sensor data instead of a person?

Computers are much **better at reading and measuring quantity data** taken from sensors for the following reasons:

- More reliable** ➔ Computers **will not forget** to take the readings.
- Accuracy** ➔ The computer will read the data more **accurately** than a person.
- Response time** ➔ Computers can **react much quicker to data received**. For example, if a patient's heart rate drops to critical levels an alarm could be sounded automatically.
- Work longer** ➔ Computers can **read data all day, every day** without getting bored or tired.
- Frequency of readings** ➔ Computers can **read the data more frequently** than humans can (1000's of times a second if needed).
- Automatic Readings** ➔ No need for a human to be present. This frees people up to carry out other tasks.
- Safer** ➔ Some **environments** can be **lethal to humans**. For example monitoring radiation levels in a damaged power station.



Some environments are **too dangerous for humans** to take regular readings (arctic for example).

Examples of Computer Measurement Applications



Key Words:

Sensors, Log, Analyse,
Analogue to Digital Conversion

The following examples show you how sensors and computers can be used to measure data:

1 Scientific Experiments

- # Data logging can be setup to automatically monitor and record data generated by scientific experiments.

Water Temperature Example:

Imagine you are conducting an experiment that records the temperature of water as it transforms from solid ice into hot steam. You would need to use the following:

- **Thermostat** - to measure the temperature of the water
- **ADC** - to convert the analogue temperature data into digital
- **Computer** - to receive and log the digital temperature data
- **Software** - to analyse the received data

How the experiment data would be logged

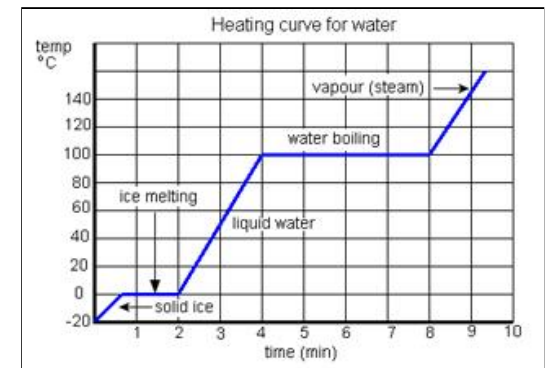
- 1 The **thermostat** would be used to **measure the constantly changing analogue temperatures** of the water.
- 2 The **ADC** is used to **convert the analogue temperature data into digital** (1's and 0's) which are then fed into the computer.
- 3 The computer and special software would then **log the constantly changing temperature data** and **automatically analyse** it in the form of charts.

NOTE!

The sensor would need to be told at **what intervals to take temperature measurements**. In our example the measurements are taken every minute.



Sensor measuring the temperature of ice



Computer software could **automatically analyse the data** and produce charts.

2 Electronic Timing

Computerised data logging can be used to accurately measure time.

Olympic 100 Metres Race Example:

In sprinting, generating accurate race times is important as, sometimes, first and second place can be separated by 1000's of a second. To accurately measure athlete's race times you would need:

- **Sound Sensor** - to record when the race is started
- **Infrared Sensor** - to record when the athlete crosses the finish line
- **ADC** - to convert the analogue data from the sensors into digital
- **Computer and Software** - to receive and log the digital data and calculate race time.

How the race time data would be logged

- 1** Race starter **fires a starting pistol** which tells each athlete that it is time to run. **Sound from the pistol** is **sent to the computer** and converted via ADC.
- 2** Sound sensor in the computer takes the signal from the starting pistol and computer **starts each athlete's timer at exactly the same time**.
- 3** The infra-red sensor projects a **beam across the finish line** which is **broken** as each athlete passes through.
- 4** The instant that the infra-red beam is broken by each athlete, a **signal is sent to the computer** which then **stops the timer**.
- 5** Software **analyses** all of the **individual athletes timings** and sorts them in **order of fastest to slowest**. This information is displayed on a board.

[Click for Picture Example](#)



Sprinters are separated by 1000's of a second and so timings have to be very accurate. [Click to see race](#)



Sensors and computers can be used alongside **digital cameras** which take up to **2000 images per second**. This helps to accurately decide who crossed the finish line first.

3 Environmental Monitoring

Environmental monitoring is where data is collected to be used for purposes such as:

- **Weather forecasting**
- **Water quality in rivers and streams**
- **Levels of air pollution**
- **etc.....**

Weather Forecasting Example:

Weather stations contain lots of sensors which are used to gather a variety of different data. For example:

- **Thermometer** - to measure temperatures
- **Humidity Sensor** - to measure the amount of water in the air
- **Barometer** - to measure air pressure
- **Anemometer** - to measure wind speed
- **Rain Gauge** - to measure the amount of rain fall over a certain period.

These sensors would feed data into a computer which is then **analysed by meteorologists** (weather forecasters). The data allows the weather forecasters to make **predictions about future weather**.



Weather stations monitor **temperature, wind speed/direction, air pressure and humidity**.

Thermometer



Humidity Sensor



Barometer



Anemometer



Rain Gauge

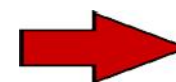
**NOTE!**

Weather station sensors **constantly collect this data**. 24 hours a day, 7 days a week, 365 days a year.



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